📄 DOCUMENT 4

Infrastructure & Technology Requirements for Phase 1 Implementation

Hawassa MESOB One-Stop Service Center

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PART 1

(Sections 1–4: Infrastructure Philosophy, Overall Technology Stack, Application Layer Architecture, Integration Layer Requirements)

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1. INFRASTRUCTURE STRATEGY FOR PHASE 1

1.1 Foundational Principle

Phase 1 must be:

- Scalable
- Secure
- Modular
- Vendor-neutral (even if Ethio Telecom is primary provider)
- API-first

The infrastructure must support:

✓ 47 Phase 1 services
✓ Multi-bureau integration
✓ Payment processing
✓ Real-time dashboards
✓ SLA monitoring
✓ Audit logging

It must also be expandable for Phase 2.

1.2 Design Philosophy

The architecture will follow a layered model:

1. Presentation Layer (Citizen interface & service counters)
2. Application Layer (Service logic & workflow engines)
3. Integration Layer (API gateway & mesh connectivity)
4. Data Layer (Databases & storage)
5. Infrastructure Layer (Servers, hosting, networking)
6. Security Layer (Identity, encryption, monitoring)

Each layer must be independently scalable.

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2. OVERALL PHASE 1 TECHNOLOGY STACK

Phase 1 requires the following core systems:

2.1 Core Systems Overview

Layer	System Component
Frontend	Service Portal & Counter Interface
Queue	Digital Queue Management System
Workflow	Service Workflow Engine
Integration	API Gateway
Data	Central Service Database

Layer	System Component
Reporting	Executive Dashboard
Payments	Payment Gateway Integration
Identity	Authentication & Role Management
Logging	Audit & Monitoring System
Hosting	Cloud or Data Center Infrastructure

2.2 Hosting Strategy (Critical Decision)

Two possible hosting models:

Option A: Ethio Telecom Hosting (Managed Infrastructure)

Pros:

- Faster deployment
- Integrated connectivity
- Single vendor accountability

Cons:

- Vendor dependency
- Long-term cost risk
- Less infrastructure control

Option B: Hybrid Hosting (City-controlled + Telecom Network)

Pros:

- Greater data sovereignty
- Vendor diversification
- Flexible scalability

Cons:

- Higher initial complexity
- Requires internal IT governance capacity

Recommendation:

Phase 1 may use managed hosting,
but contract must guarantee:

- API control
- Data ownership
- Migration rights
- Exit clause protections

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3. APPLICATION LAYER REQUIREMENTS

This layer runs the 47 services.

3.1 Service Management System (Core Engine)

Must support:

- Multi-service configuration
- SLA timers
- Service status tracking
- Bureau routing
- Digital approval workflows
- Document upload
- Digital signature capability

Capacity Requirement:

- Minimum 200 concurrent users (scalable to 500)
- 5,000–10,000 transactions per day capacity

3.2 Queue Management System

If provided by Ethio Telecom:

Must integrate with:

- Service selection

- Ticket issuance
- Counter assignment
- Performance analytics
- SMS notification

Critical:

Queue system must not operate in isolation.
It must link with service workflow system.

3.3 Payment Processing Integration

Must support:

- Cash (counter entry)
- POS terminals
- Mobile payment (if available)
- Bank integration

Required capabilities:

- Real-time payment confirmation
- Automatic revenue posting
- Daily reconciliation report
- Receipt generation with QR verification

3.4 Executive Dashboard

For Mayor & Bureau Heads.

Must display:

- Daily revenue
- Service volume
- SLA compliance rate
- Pending approvals
- Escalations
- Bureau performance comparison

Dashboard must update in real time.

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4. INTEGRATION LAYER REQUIREMENTS

This is the most critical layer.

Without proper integration, MESOB becomes only a digital front desk.

4.1 API Gateway Requirement

Required Capabilities:

- Central API management
- Authentication & authorization
- Rate limiting
- Logging
- Error monitoring
- Version control

It must connect to:

- Land Bureau systems
- Revenue Bureau systems
- Trade & Industry systems
- Investment systems
- Health & Education systems

4.2 Bureau API Endpoint Scenarios

Two categories exist:

Category A: Bureaus With Existing IT Systems

Example:

- Land registry database

- Tax system

Required:

- API wrapper layer
- Secure authentication
- Data mapping
- Field validation

Responsibility Question:

Who builds the API wrapper?

This must be contractually clarified with Ethio Telecom.

Category B: Bureaus Without IT Systems

For these bureaus:

- Backend system must be built
- Database must be created
- API endpoints must be designed
- Basic workflow logic must be developed

Critical Issue:

If Ethio Telecom only provides infrastructure,
then API development responsibility must be:

Either:

- MESOB system vendor
  Or:
- Separate integration contractor

This cannot remain undefined.

4.3 API Design Standards

All APIs must follow:

- REST-based architecture

- JSON data format
- Token-based authentication
- Standard error handling
- Audit log generation

Mandatory:

Each API must have:

- Technical documentation
- Test environment
- Version tracking

4.4 Data Exchange Mesh Alignment

National MESOB taskforce uses:

API Gateway / Mesh Architecture

Hawassa MESOB must:

- Connect upward to federal gateway
- Maintain local API registry
- Avoid duplicate integration

Clear boundary must be defined:

Local vs National integration scope.

END OF DOCUMENT 4 – PART 1

Covered:

✓ Infrastructure design philosophy
✓ Overall technology stack
✓ Application layer requirements
✓ Integration layer requirements
✓ Bureau endpoint scenarios

Reply with:

“Proceed with Document 4, Part 2.”

Next we will produce:

Data layer
Security architecture
Network infrastructure
Capacity sizing
Disaster recovery
Monitoring & maintenance requirements.

📄 DOCUMENT 4

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PART 2
(Sections 5–9: Data Layer, Security Architecture, Network Infrastructure, Capacity Sizing, Disaster Recovery & Monitoring)

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5. DATA LAYER REQUIREMENTS

The data layer is the backbone of MESOB.

Without structured, clean, secure data:

- SLA tracking fails
- Revenue reconciliation fails
- Audit traceability collapses
- Integration breaks

5.1 Core Databases Required

Phase 1 requires:

1. Central Service Transaction Database

Stores:

- All 47 service requests
- Status
- Bureau routing history
- SLA timers
- Approval decisions
- Payment linkage

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2. Revenue & Payment Ledger Database

Stores:

- Payment transactions
- Revenue codes
- Receipt IDs
- Reconciliation records
- Daily settlement logs

Must support financial audit standards.

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3. Bureau Master Data Registry

Stores:

- Service definitions
- Bureau routing rules
- Approval authority matrix
- SLA configurations

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4. User & Identity Database

Stores:

- User roles
- Bureau permissions
- Counter staff access
- System administrator rights
- Login audit logs

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5. Document Storage System

Stores:

- Uploaded documents
- Lease files
- Business certificates

- Inspection reports
- Scanned historical records

Minimum requirement:

- Structured document index
- Secure access control
- Version history
- Retention policy configuration

5.2 Data Ownership & Governance

Critical principle:

All MESOB data belongs to Hawassa City Administration.

Contract with vendor (e.g., Ethio Telecom) must ensure:

- Full data export rights
- No proprietary data lock-in
- Standard database formats
- Backup access for city IT team

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6. SECURITY ARCHITECTURE

Security is non-negotiable.

MESOB handles:

- Land ownership data
- Tax records
- Identity records
- Investment data

Security architecture must include:

6.1 Identity & Access Management (IAM)

Must support:

- Role-based access control (RBAC)
- Multi-level authorization
- Segregation of duties
- Password policy enforcement
- Login audit trail

Example:

Land officer cannot access health records.

Revenue clerk cannot modify land ownership data.

6.2 Authentication Protocol

Minimum requirements:

- Token-based authentication
- Encrypted communication (HTTPS/TLS)
- Session timeout control
- Account lockout policy

6.3 Data Encryption

Encryption must apply:

- Data in transit
- Sensitive data at rest
- Backup files

Critical:

Land and revenue data must be encrypted at database level.

6.4 Audit Logging & Monitoring

System must log:

- Login attempts
- Data modification
- Approval decisions
- Payment overrides
- API calls

Logs must be:

- Immutable
- Timestamped
- Exportable for audit review

6.5 Fraud & Abuse Controls

Required controls:

- Duplicate transaction detection
- Revenue anomaly alerts
- SLA manipulation alerts
- Administrative override logging

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7. NETWORK INFRASTRUCTURE REQUIREMENTS

MESOB physical building must support:

- High-speed internet connectivity
- Secure LAN architecture
- Counter-to-server communication
- Backup connectivity

7.1 Connectivity Requirements

Minimum:

- Primary high-speed fiber link
- Secondary backup connection

- Redundant routing configuration

If Ethio Telecom is primary provider:

Contract must guarantee:

- 99.5% uptime minimum
- Response time SLA
- Fault escalation timeline

7.2 Internal Network Architecture

Required components:

- Core switch
- Access switches
- Secure Wi-Fi (staff only)
- Segmented VLAN architecture

Segmentation example:

- Counter network
- Admin network
- Server network
- Guest network (if public Wi-Fi offered)

7.3 Hardware Requirements (Logical Level)

Minimum required:

- Application servers
- Database servers
- Storage system
- Firewall appliance
- Backup appliance
- POS terminals
- Counter PCs
- Queue display screens
- UPS systems

(Exact quantities detailed in Document 8 – BoQ)

8. CAPACITY SIZING FOR PHASE 1

Phase 1 projected load:

- 1,000–2,500 daily transactions
- 50–70 concurrent counter users
- 100+ internal bureau users
- 5,000–10,000 daily API calls

Infrastructure must support:

- 3-year growth projection
- 30–50% capacity buffer

8.1 Storage Estimate

Estimated minimum storage:

- 3–5 TB structured data
- 10–15 TB document storage
- Expandable architecture

8.2 Compute Estimate

Must support:

- Workflow processing
- Payment integration
- Dashboard analytics
- API traffic handling

Virtualized or cloud-managed servers preferred.

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9. DISASTER RECOVERY & BUSINESS CONTINUITY

Critical for government services.

9.1 Backup Strategy

Must include:

- Daily incremental backup
- Weekly full backup
- Monthly archival snapshot
- Off-site storage replication

9.2 Disaster Recovery Model

Recommended:

- Secondary data center or cloud failover
- Recovery Time Objective (RTO): < 4 hours
- Recovery Point Objective (RPO): < 1 hour

9.3 Power Backup

MESOB building must include:

- UPS for all critical equipment
- Generator backup
- Surge protection

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10. MONITORING & MAINTENANCE

Continuous monitoring required for:

- Server health
- API performance
- Network stability
- Storage utilization
- SLA compliance

Must include:

- Real-time alerting
- Performance dashboard
- Incident ticketing system

Vendor must provide:

- 24/7 support
- Defined escalation matrix
- Monthly performance report

PHASE 1 INFRASTRUCTURE READINESS CHECKLIST

Before launch, confirm:

✓ API endpoints operational
✓ Hosting environment tested
✓ Backup system validated
✓ Security audit completed
✓ Load testing conducted
✓ Bureau access configured
✓ Payment reconciliation tested

DOCUMENT 4 COMPLETE

This concludes:

✓ Full technology stack
✓ Data layer
✓ Security architecture

✓ Network design
✓ Capacity planning
✓ Disaster recovery
✓ Monitoring requirements

Reply with:

“Proceed with Document 5, Part 1.”

Next we begin:

Stakeholder Mapping & Bureau Readiness Checklists
(With explicit separation between bureaus with IT systems vs those without systems, including API responsibility clarification).